Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Nayagam, Vedha Ph.D.		
Project Title:	Stretched Diffusion Flames in Von Karman Swirling Flows		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	v.nayagam@grc.nasa.gov	**Fax:**	FY 216 433-3793
PI Organization Type:	UNIVERSITY	**Phone:**	216 433-8702
Organization Name:	National Center for Microgravity Research		
PI Address 1:	Mail Stop 110-3, National Center for Microgravity Combustion		
PI Address 2:			
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:	Closed 2/4/02		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	01/22/2001	**End Date:**	01/22/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Nayagam, Vedha		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The von Karman swirling flows are generated by spinning a fuel disk at different speeds. This geometry provides a unique way of establishing a desired stretch rate from very low to very high values. Also, the governing equation for this geometry reduces to an one-dimensional problem making it possible to include complex chemical and transport mechanisms. Also, close to the extinction boundaries the diffusion flame breaks symmetry and forms complex dynamic patterns some of which are similar to the patterns observed in biological and chemical systems. We examine pattern formation along the flammability boundaries using a gas-fueled, spinning porous-disk burner. Also, theoretical descriptions of these edge-flames are developed using activation energy asymptotic models. The objective of this work is to investigate, both experimentally and theoretically, the effect of stretch on diffusion flames embedded in swirling flows. Theoretical models using asymptotic techniques (activation energy and rate ratio asymptotics) will be developed to interpret the experimental observations. Both the experiments and the theoretical models will be focused on the flammability boundaries in the low stretch and high stretch regimes. We have also		

Task Description:	discovered the existence of pattern formation in diffusion flames in von Karman swirling flows close to the extinction boundaries that exhibit many similarities to the patterns observed in biological systems. One of our objectives to understand the commonalities between these phenomena. The motivation for this study comes from the need to understand material flammability conditions in the slow convective environments commonly encountered in spacecraft environments. Here, flames that would normally extinguish on earth may be sustained. It is also hoped that the results obtained from studies of laminar-flame extinction may be applied for predictions in turbulent situations through estimates of strain rates in turbulent fields. Also, the pattern formation phenomena in diffusion flames was discovered during our previous NASA funded grant period. In the current study we further investigate this phenomena using a newly designed gas-fueled, spinning-disk burner. The rotating spiral flames observed in these experiments mimic electrical pulse propagation in heart tissues and exhibit similar dynamics. We hope to model these flame dynamics and try to identify the underlying commonalities behind these phenomena.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Nayagam V, Williams FA. "Curvature Effects on Edge-Flame Propagation in the Pre-mixed-Flame Regime." Combustion Science and Technology. 2004;176(12):2125-42. https://doi.org/10.1080/00102200490514976 , Jun-2004